UiA Fakultet for teknologi og realfag

English

Solutions: MA-224 – 4. December 2024

Problem 1

Find the flaws in the following answer and proof, and fix them.

Question: For which natural numbers n is $n + n = n \cdot n$?

Answer: $n = 2$

Proof:

$n + n = n \cdot n$	`given`
$n \cdot 2 = n \cdot n$	`definition of multiplication`
$2 \cdot n = n \cdot n$	`commutativity of` $\cdot$
$2 = n$	`division by n`
$n = 2$	`commutativity of` $=$

Solution:

The answer is incomplete, because this is a quadratic equation which has two solutions. The second possible solution is $n = 0$. The problem with the proof is that we cannot divide by n if $n = 0$. This leads to the following corrected result.

Answer: $n = 0 \vee n = 2$

Proof:

$n + n = n \cdot n$	`given`
$n \cdot 2 = n \cdot n$	`definition of multiplication`
$\textit{case } 1 : n = 0$	
$0 \cdot 2 = 0 \cdot 0$	`because` $n = 0$
$0 = 0$	`definition of multiplication`
$\textit{true}$	`definition of equality`
	`this means n = 0 is a solution`
$\textit{case } 2 : n \neq 0$	
$2 \cdot n = n \cdot n$	`commutativity of` $\cdot$
$2 = n$	`division by n`
$n = 2$	`commutativity of` $=$

Problem 2

Prove the following theorem by induction using the definitions of Peano addition.
ThEx: $\forall n \in \mathbb{N} : n + 1 = s(n)$

For your convenience, here are the definitions of Peano addition.

+.1: $\forall a \in \mathbb{N} : 0 + a = a$

+.2: $\forall a \in \mathbb{N} : \forall x \in \mathbb{N} : s(x) + a = s(x + a)$

n.1: $1 = s(0)$

Solution:

The proof is done inductively.

(1) Induction start: Prove for $n = 0$.

$$\begin{aligned} 0 + 1 &= 1 && \texttt{by +.1} \\ &= s(0) && \texttt{by n.1} \end{aligned}$$

(2) Induction assumption: for a Peano number $n = k$.
$k + 1 = s(k)$

(3) Induction step: Prove for $n = s(k)$.

$$\begin{aligned} s(k) + 1 &= s(k + 1) && \texttt{by +.2} \\ &= s(s(k)) && \texttt{by induction assumption} \end{aligned}$$

Problem 3

Consider the following three derivations.

(1) $E \to L = R \to L = 3 \to L + L = 3 \to 1 + L = 3 \to 1 + 2 = 3$

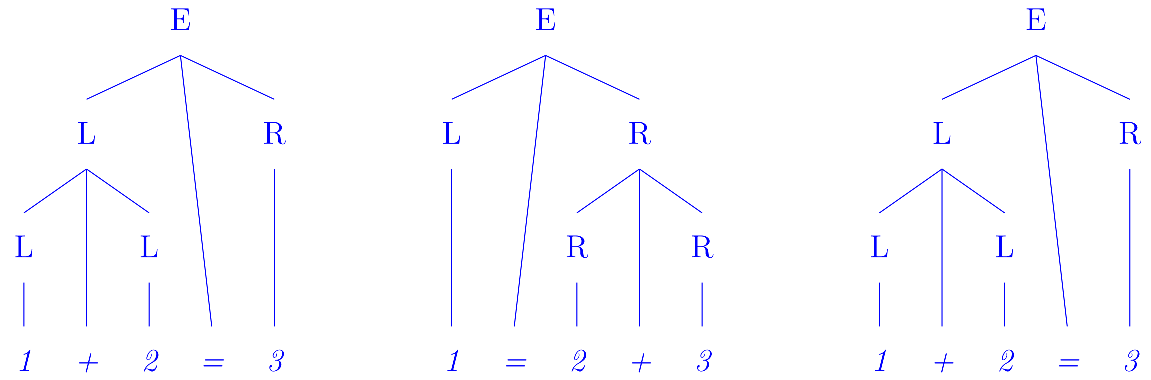
(2) $E \to L = R \to 1 = R \to 1 = R + R \to 1 = 2 + R \to 1 = 2 + 3$

(3) $E \to L = R \to L + L = R \to 1 + L = R \to 1 + 2 = R \to 1 + 2 = 3$

Discuss the ambiguity of the grammar that allows those derivations. Give reasons.

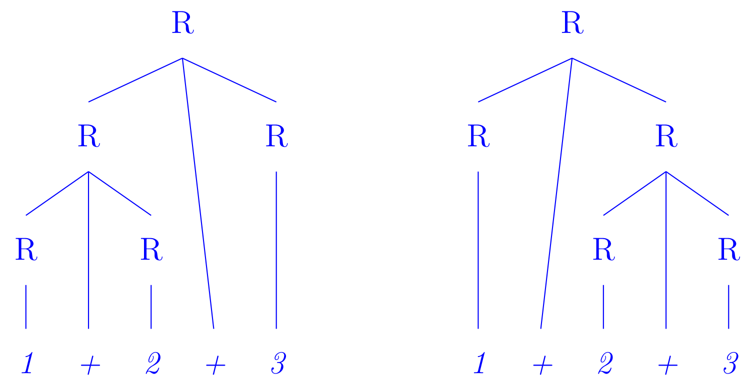
Solution:

First, we look at the syntax trees for the three derivations.



The tree for the first and the last derivation are the same and do not mean ambiguity. The tree for the second derivation is different, but it is also a different string, such that this again does not mean any ambiguity.

However, the derivations show that there must be a rule $R \to R + R$ and a rule $L \to L + L$. Each of these two rules creates an ambiguity for a string like 1+2+3 with the following two trees for, e.g., R.



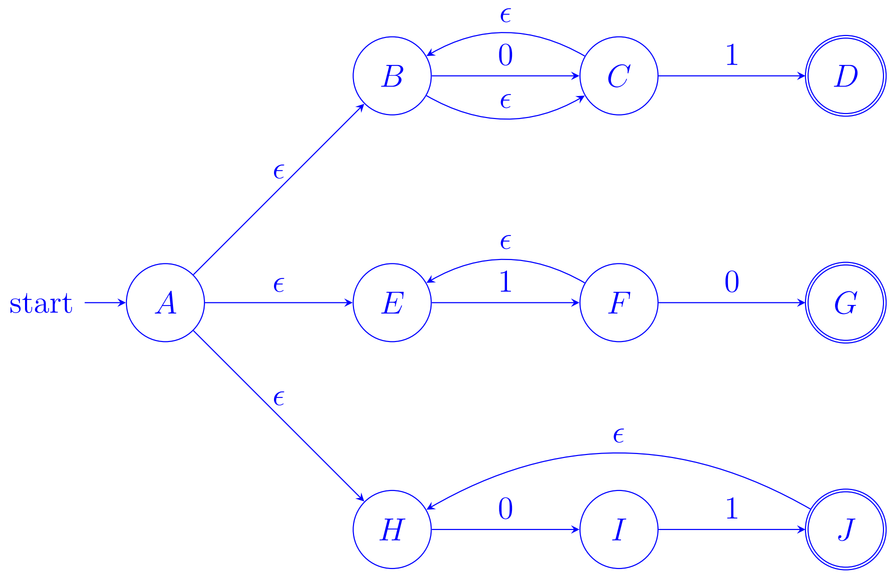
Therefore, any grammar adhering to the given derivations must be ambiguous.

Problem 4

Translate the regular expression 0*1|1?0|(01)+ to a right-regular grammar.
Write the grammar as a 4-tuple (Terminals, NonTerminals, startSymbol, Rules).
Remember that a right-regular grammar has only rules of the form (A, t), (A, tB), (A, B), or (A,), where t is a terminal and A, B are nonterminals.

Solution:

There are several possible ways to do this translation. We follow the approach from the lecture. First, we translate the regular grammar to an NFA.



Now we translate the NFA into a grammar G.

```
G=( {0,1},
    {A,B,C,D,E,F,G,H,I,J},
    A,
    { (A,B), (A,E), (A,H), (B,0C), (B,C), (C,1D), (C,B), (D,),
      (E,1F), (F,E), (F,0G), (G,), (H,0I), (I,1J), (J,), (J,H)
    }
  )
```

Problem 5

Extend the JML definition of Peano below with a function called mod2, which calculates (p % 2) for a Peano number p. Here, (p % 2) is the remainder when dividing p by 2.

The recursive mathematical definition of mod2 is as follows.

```
for all X in Peano
  mod2(0)       = 0
  mod2(s(0))    = s(0)
  mod2(s(s(X))) = mod2(X)
```

Here is the partial specification of Peano.

```
public interface Peano {
  //@ ensures \result != null
  public Peano zero();
```

```
  //@ requires p != null;
  //@ ensures \result != null && \result != zero()
  public Peano succ(Peano p);

  //@ ensures \result == succ(zero)
  public Peano one();

  //@ requires p != null && p != zero();
  //@ ensures \result != null && succ(\result) == p
  public Peano pred(Peano p);

  //@ requires p1 != null && p2 != null;
  //@ ensures \result == (p2==zero()) ? p1 : succ(add(p1,pred(p2)))
  public Peano add(Peano p1, Peano p2);
}
```

Solution:

We can extend the JML as follows, showing only the relevant new part of the specification.

```
public interface Peano {
...
  //@ requires p != null;
  //@ ensures \result == (p==zero()) ? zero()
  //                        : (p==one()) ? one()
  //                             : mod2(pred(pred(p)))
  public Peano mod2(Peano p);
}
```

Problem 6

The relation $a|b$ on natural numbers (read "a divides b" or "b is a multiple of a") is true iff there is a natural number n such that $a \cdot n = b$.

$a|b \equiv \exists n \in \mathbb{N} : a \cdot n = b$

(a) Is $|$ reflexive?

(b) Is $|$ symmetric?

(c) Is $|$ anti-symmetric?

(d) Is $|$ transitive?

(e) Is $|$ an equivalence relation, partial order, or total order?

Give reasons.

Hint: Be careful with 0.

Solution:

(a) For all natural numbers a it is true that $a \cdot 1 = a$, which implies $a|a$. Therefore, $|$ is reflexive.

(b) We know that $2|4$, but not $4|2$. Therefore, $|$ is not symmetric.

(c) Let $a|b$ and $b|a$. This means there are n and m such that $a \cdot n = b$ and $b \cdot m = a$. By replacing the second equation in the first one we get $b \cdot m \cdot n = b$.

We consider two cases.

Case 1) $b = 0$

Then $a = b \cdot m = 0 \cdot m = 0$, and hence $a = b$.

Case 2) $b \neq 0$

Now we can divide the equation $b \cdot m \cdot n = b$ by b and we get $1 \cdot m \cdot n = m \cdot n = 1$. There is only one solution for n and m for this equation, which is $n = m = 1$. Therefore, $a = b \cdot m = b \cdot 1 = b$.

In both cases, we concluded with $a = b$. Therefore, $|$ is antisymmetric.

(d) Let $a|b$ and $b|c$. This means there are n and m such that $a \cdot n = b$ and $b \cdot m = c$. By replacing the first equation in the second one we get $a \cdot n \cdot m = c$. This means that there is a natural number $k = n \cdot m$, such that $a \cdot k = c$, and hence $a|c$.

Therefore, $|$ is transitive.

(e) $|$ is not an equivalence relation because it is not symmetric.

$|$ is a partial order because it is reflexive, antisymmetric, and transitive.

$|$ is not a total order because neither $2|3$ nor $3|2$.

Problem 7

Find the solution to the initial value problem

$$\begin{cases} a_n - 4a_{n-1} + 4a_{n-2} = 0 \\ a_0 = 0 \\ a_1 = 1\,. \end{cases}$$

Solution:

The characteristic equation associated to this recurrence relation is

$$r^2 - 4r + 4 = 0$$

and has a single real solution $r = 2$. Therefore the general solution to the recurrence relation is

$$a_n = A \cdot 2^n + B \cdot n \cdot 2^n\,.$$

To find the particular solution, we apply the initial values condition and get $0 = a_0 = A + 0 \cdot B \Rightarrow A = 0$, and $1 = a_1 = 0 \cdot 2^1 + B \cdot 1 \cdot 2^1 = 2B \Rightarrow B = 1/2$. The particular solution is therefore

$$a_n = \frac{1}{2} \cdot n \cdot 2^n = n \cdot 2^{n-1}\,.$$

Problem 8

Let $(G, \cdot)$ be a group with neutral element $e \in G$. For each element $x \in G$, we write x^{-1} for the inverse of x.

Let $a,b \in G$.

(a) Show that $(a \cdot b)^{-1} = b^{-1} \cdot a^{-1}$.

(b) Simplify the expression $a \cdot (b^{-1} \cdot a)^{-1}$ as much as possible.

Solution:

(a) Let $y = b^{-1} \cdot a^{-1}$. Then

$$y \cdot (a \cdot b) = (b^{-1} \cdot a^{-1}) \cdot (a \cdot b) = (b^{-1} \cdot (a^{-1} \cdot a)) \cdot b) = (b^{-1} \cdot e) \cdot b) = b^{-1} \cdot b) = e$$

which shows that $y = (a \cdot b)^{-1}$.

(b)
$$a \cdot (b^{-1} \cdot a)^{-1} = a \cdot (a^{-1} \cdot (b^{-1})^{-1}) = (a \cdot a^{-1}) \cdot (b^{-1})^{-1} = e \cdot b = b\,.$$

Problem 9

Let $\phi(n) = |P_n|$ be the Euler totient function.

(a) Calculate $\phi(n)$ for $n = 2,4,8$ and 16 by counting the elements of P_n in each case.

(b) Calculate $\phi(1024)$.

Solution:

(a)

$$\begin{aligned} |P_2| &= |\{1\}| = 1 \\ |P_4| &= |\{1,3\}| = 2 \\ |P_8| &= |\{1,3,5,7\}| = 4 \\ |P_{16}| &= |\{1,3,5,7,9,11,13,15\}| = 8\,. \end{aligned}$$

(b) The group $P_n = \{x \in \mathbb{N} \mid 1 \leq x \leq n \text{ and } \gcd(x,n) = 1\}$ consists of the natural numbers between 1 and n which are coprime to n. When $n = 2^k$, for $k \geq 1$, this means all natural numbers n less than 2^k which are coprime to 2, i.e. each odd number $(2r-1)$ from $r = 1$ to $r = 2^{k-1}$. Therefore $\phi(1024) = 512$.

Problem 10

Let $(G, \cdot)$ be a finite group with neutral element $e \in G$. Assume that there exists an element $g \in G$ such that $g \neq e$ and $g \cdot g = e$. Prove that the order of G is an even number.

Solution:

By the assumption, $H = \{e,g\}$ is a cyclic subgroup of order 2. By Lagrange's theorem, $|G|$ is divisible by 2, hence even.

Problem 11

We want to create an encoding function

$$E : (\mathbb{Z}/2)^{\times 3} \to (\mathbb{Z}/2)^{\times 6}$$

given by multiplication with a generator matrix:

$$E(w) = w \cdot G\,.$$

The encoded messages will be transmitted over a noisy channel.

(a) Find a generator matrix G such that the associated encoding E is capable of correcting all transmission errors of weight 1. Prove that G satisfies this property.

(b) Based on the matrix G you chose in the answer to the question above, compute the syndrome of the bitstring $r = (101001)$.

Solution:

(a) Let

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

We compute the weigths of the non-zero codewords:

$$\begin{aligned}
[001] \cdot G &= [001111] \\
[010] \cdot G &= [010101] \\
[011] \cdot G &= [011010] \\
[100] \cdot G &= [100110] \\
[101] \cdot G &= [101001] \\
[110] \cdot G &= [110011] \\
[111] \cdot G &= [111100]
\end{aligned}$$

Since the encoding is a group code, the minimum distance between codewords, l_{min}, is equal to the minimum weight of the non-zero codewords. From the list above we see that $l_{min} = 3$. All transmission errors of weight $(3-1)/2 = 1$ are correctable.

(b) The parity check matrix H is

$$H = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

The syndrome is then given by

$$H \cdot r^T = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Problem 12

The encoding function

$$E : (\mathbb{Z}/2)^{\times 3} \to (\mathbb{Z}/2)^{\times 5}$$

is given by matrix multiplication $E(w) = w \cdot G$ with some (3×5)-generator matrix $G = [I_3\ A]$.

Prove, using the theory of *decoding by coset leaders*, that by using the encoding E we can not correct all transmission errors of Hamming weight 1.

Hint: Use Lagrange's theorem to compute the number of cosets of $C \subset (\mathbb{Z}/2)^{\times 5}$, where C is the subgroup of codewords for E.

Solution:

The group $(\mathbb{Z}/2)^{\times 5}$ has order $2^5 = 32$ and C has order $2^3 = 8$. Lagrange's theorem tells us that $[(\mathbb{Z}/2)^{\times 5} : C] = 32/8 = 4$. However, there are 5 different bitstrings of Hamming weight 1, so there must exist at least two error patterns $e \neq e'$ such that $wt(e) = wt(e') = 1$ and for which the cosets $e + C = e' + C$ are equal. In other words, there exists codewords $c \neq c'$, and error patterns $e \neq e'$ of weight 1 such that $c + e = c' + e'$. This is an example of an uncorrectable 1-bit error.